# SCOREBOARD – Hadamard 14160 Paley-I (2026-10-08)

Cutoff: 2026-10-09T03:18:04Z (~11:18PM ET)

| Object | Status | max\|HHˆT-nI\| | Wall (s) | SPK |
|---|---|---|---|---|
| H_14160 Paley-I (q=14159) | PASS | 0 | 2671.8467 | **IN SPK** (Paley-I) |
| H_14160 Paley-II (q=7079) | FAIL/SKIP | – | – | q=7079 prime but ===3 mod 4 needs ===1 |
| H_14160 Kronecker (4*3540) | SKIP deferred | – | – | 3540===0 mod 4 order-admissible; H_3540 via Paley-I q=3539 ALSO ELIGIBLE – deferred |

Units: 662/662 * row blocks 111/111 * overall_pass=True * wall_total=2712.4971s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
**IN SPK via Paley-I**; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884+ retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ271*317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7*167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*3*7*167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2ˆ2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3ˆ2*1571 NOT prime -> FAIL;**

Kronecker 14140=4*3535; 3535=57101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2^257101 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | FIRST Paley-II DIRECT after 14160: q2=7109->n=14220 (q2=7109 prime ===1 mod 4; Paley-I q=14219=59241 NOT prime -> FAIL; Kronecker 14220=43555; 3555=3^2579 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^23^2579 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); also later Paley-I 14208 q=14207 before that Paley-II) | Kronecker-only (no direct Paley route; note-only, NOT on box): 14192=43548 (3548=2^2887 ===0 mod 4; H_3548 via Paley-I q=3547 prime ===3 mod 4); 14224=43556 (3556=2^27127 ===0 mod 4; H_3556 via Paley-II q2=1777 prime ===1 mod 4); 14240=43560 (3560=2^3589 ===0 mod 4; H_3560 via Paley-I q=3559 prime ===3 mod 4); 14112=43528 (3528=$2^{3^{3^{\wedge}2}7}$2 ===0 mod 4; H_3528 via Paley-I q=3527 prime ===3 mod 4) remains below-after_n historical; none other in 14160..14208 with a direct H_m-only route (14176=43544 no direct H_3544) | next Paley-I DIRECT: q=14207->n=14208 (prime ===3 mod 4; no prime ===3 mod 4 in 14160..14206; 14173/14177/14197 ===1 skipped; FIRST n===0 mod 4 after 14160 with Paley-I DIRECT; Paley-II q=7103 prime but ===3 mod 4 (needs ===1) -> FAIL/SKIP; Kronecker 14208=43552; 3552=2?337 ===0 mod 4 order-admissible but H_3552 has NO direct classical Paley route -> FAIL/SKIP note-only; 14208=2?337 ===0 mod 16) (or gap 2052 / Track B)